



(51) International Patent Classification:

A61K 31/4545 (2006.01) A61K 31/438 (2006.01)

A61K 31/435 (2006.01) A61P 25/06 (2006.01)

A61K 31/437 (2006.01)

(21) International Application Number:

PCT/IN2022/051108

(22) International Filing Date:

20 December 2022 (20.12.2022)

(25) Filing Language:

English

(26) Publication Language:

English

(30) Priority Data:

202141059433 20 December 2021 (20.12.2021) IN

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(81) Designated States (unless otherwise indicated, for every kind of national protection available): AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CV, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM, TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) Designated States (unless otherwise indicated, for every kind of regional protection available): ARIPO (BW, CV, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, ME, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

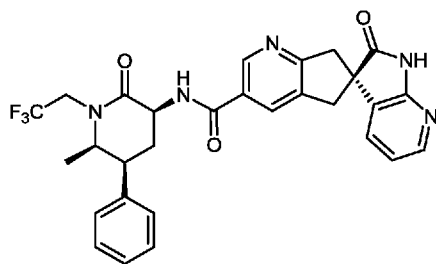
Declarations under Rule 4.17:

- as to applicant's entitlement to apply for and be granted a patent (Rule 4.17(ii))
- of inventorship (Rule 4.17(iv))

Published:

- with international search report (Art. 21(3))
- before the expiration of the time limit for amending the claims and to be republished in the event of receipt of amendments (Rule 48.2(h))

(54) Title: PROCESS FOR THE PREPARATION OF A PURE AMORPHOUS FORM OF UBROGEPANT



Formula-1.

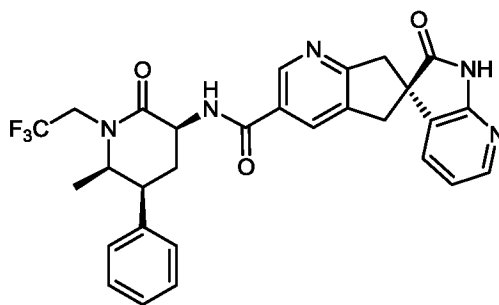
(57) Abstract: The present application relates to a pure amorphous form of Ubrogapant and its process for the preparation thereof. Ubrogapant is chemically known as (3'S)-N-((3S, 5S, 6R)-6-methyl-2-oxo-5-phenyl-1-(2, 2, 2-trifluoroethyl) piperidin-3-yl)-2'-oxo-1', 2', 5, 7-tetrahydro spiro [cyclopenta [b] pyridine-6, 3'-pyrrolo [2, 3b] pyridine]-3-carboxamide and it is represented by the following structural formula-1.

Process for the preparation of a pure amorphous form of Ubrogapant

This application claims the benefit of priority of our Indian patent application number 202141059433 filed on December 20, 2021, which is incorporated herein by reference.

Field of the invention:

The present application relates to pure amorphous form of Ubrogapant and its process for the preparation thereof. Ubrogapant represented by the following structural formula-1.



Formula-1.

Background of the invention:

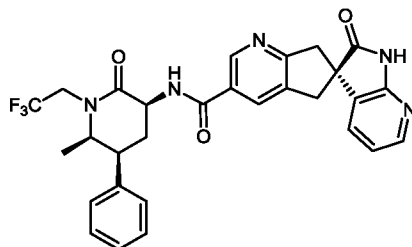
Ubrogapant formula-1 is chemically known as (3'S)-N-((3S,5S,6R)-6-methyl-2-oxo-5-phenyl-1-(2,2,2-trifluoroethyl) piperidin-3-yl)-2'-oxo-1',2',5,7-tetrahydrospiro [cyclopenta[b]pyridine-6,3'-pyrrolo[2,3-b]pyridine]-3-carboxamide, which was approved for the indication of calcitonin gene-related peptide receptor antagonist indicated for the acute treatment of migraine with or without aura in adults with the brand name of Ubrelvy®.

Ubrogapant or its pharmaceutically acceptable salts described in US8754096B2. This also discloses the process for the preparation of Ubrogapant.

US9174989B2 patent describes the various polymorphic forms of Ubrogapant such as Ubrogapant monohydrate, Ubrogapant trihydrate, Ubrogapant acetonitrile solvate, Ubrogapant acetonitrile water solvate, Ubrogapant pseudo amorphous and Ubrogapant L-tartaric acid co-crystal.

Claims:

1. A pure amorphous form of Ubrogapant of formula-1



Formula-1.

2. The pure amorphous form of Ubrogapant as claimed in claim 1, which is having a purity of greater than about 99% by HPLC.
3. The pure amorphous form of Ubrogapant as claimed in claim 1, which is essentially free of residual solvents.
4. The pure amorphous form of Ubrogapant as claimed in claim 1, which contains less than about 5% of crystalline forms of Ubrogapant.
5. The pure amorphous form of Ubrogapant as claimed in claim 1, which is stable.
6. The pure amorphous Ubrogapant as claimed in claim 5, which does not convert to any other solid forms when stored at a temperature of up to about 40° C and at a relative humidity of about 25% to about 75% for about six months or more.
7. A process for the preparation of pure amorphous form of Ubrogapant of formula-1, the process comprising the steps of:
 - a) dissolving Ubrogapant of formula-1 in a solvent,
 - b) isolating amorphous form of Ubrogapant by combining with an anti-solvent.
8. The process as claimed in claim 7, wherein dissolving Ubrogapant in step-a) is carried out at a temperature ranging from about 25°C to reflux temperature of the solvent used.
9. The process as claimed in claim 7, wherein the solvent in step-a) is selected from ester solvent, alcohol solvent, ketone solvent, nitrile solvent, polar-aprotic solvents or mixtures thereof.

(1/2)

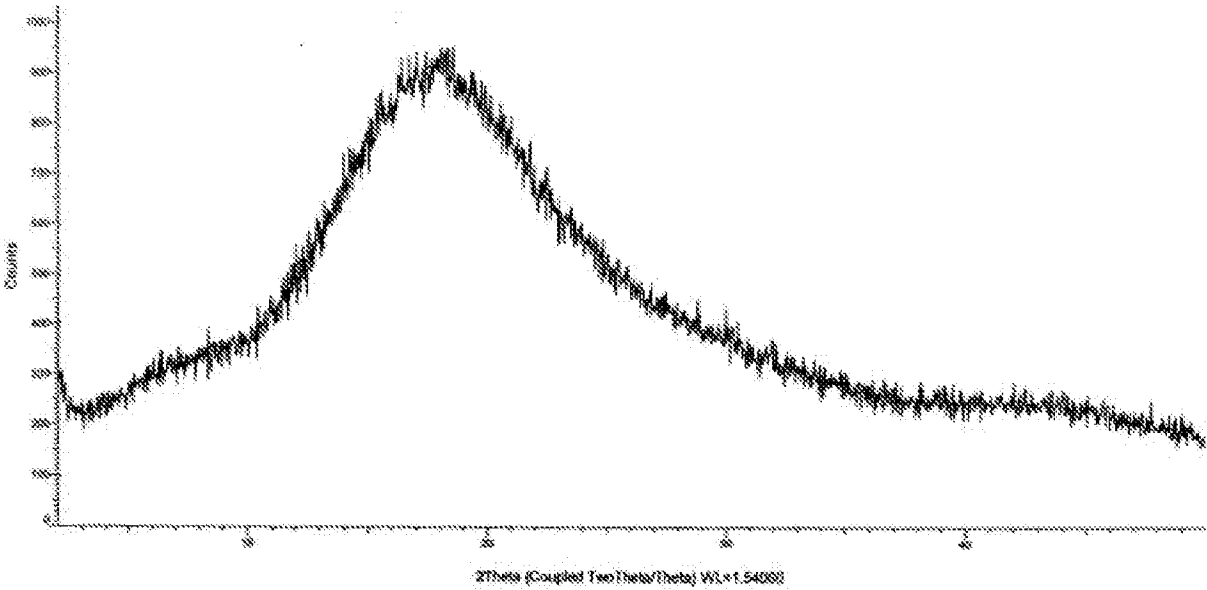


Figure-1

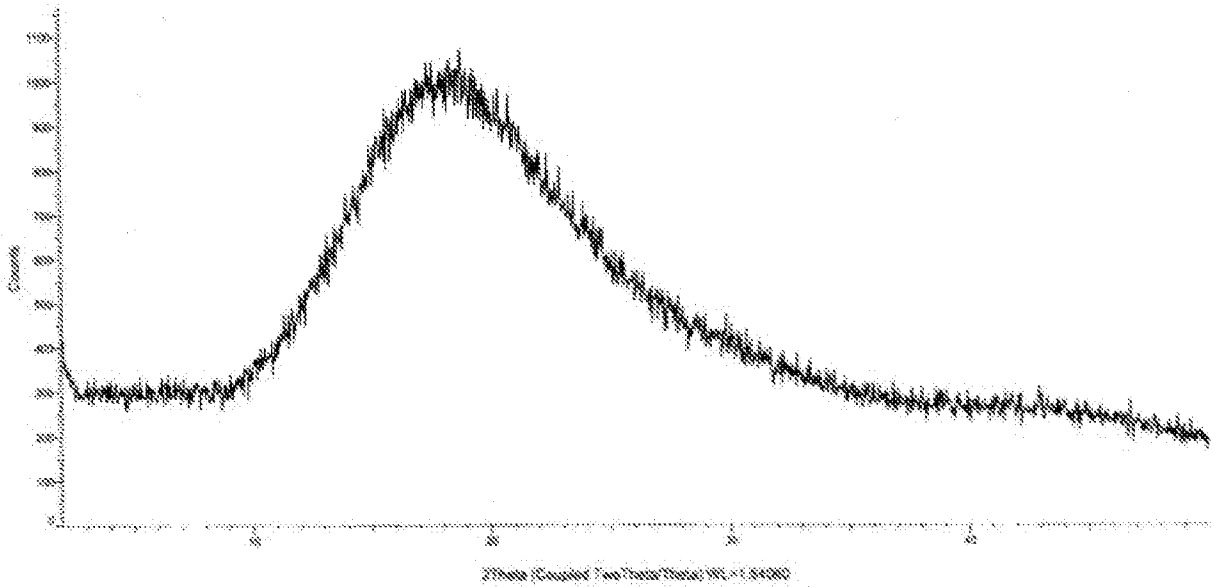


Figure-2